ShieldFi

Decentralised Insurance Pool

Blockchain Technologies 2 — Capstone Final Project

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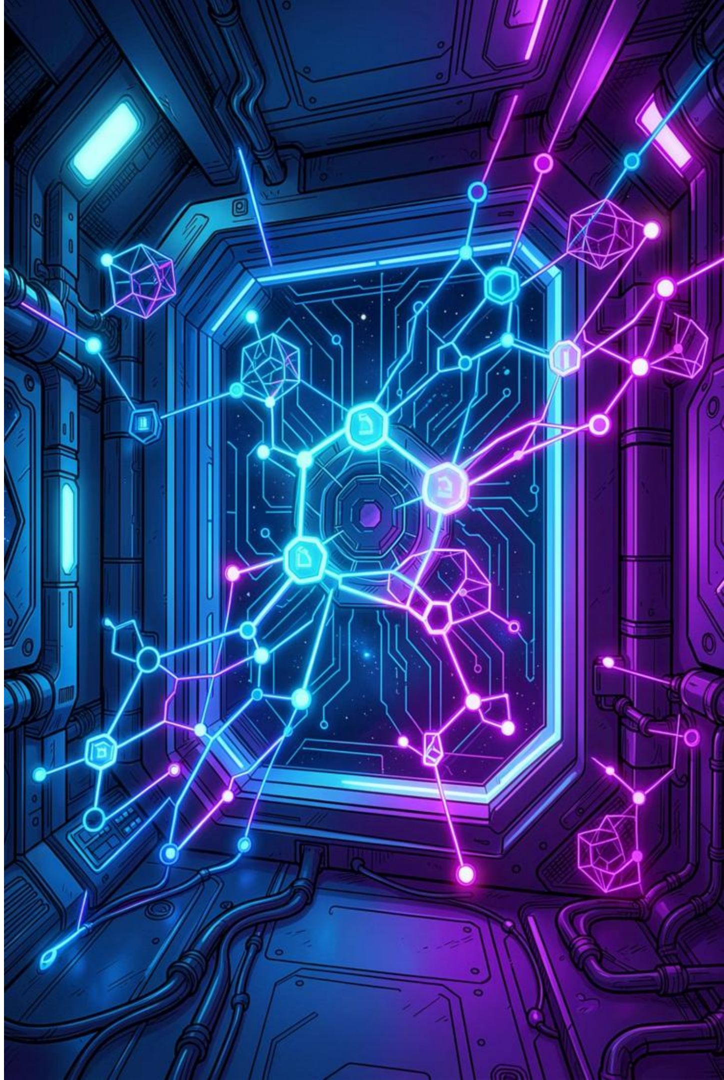


PROBLEM STATEMENT

Traditional Insurance Is Broken

Centralised insurance systems are plagued by opaque processes, slow payouts, and misaligned incentives — leaving policyholders powerless and under-served.

🏛️ Centralised Control

Single entities adjudicate claims with no on-chain accountability or community oversight.

🕵️ Zero Transparency

Policyholders cannot verify reserve adequacy, premium flows, or claims decisions.

⏳ Slow Payouts

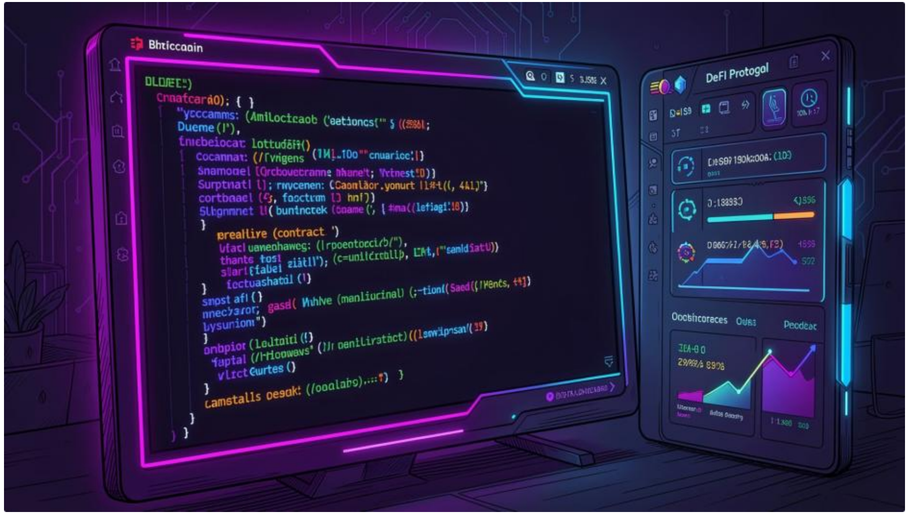
Manual review cycles create weeks-long delays — exactly when users need funds most.

⚖️ Misaligned Incentives

Insurers profit by denying claims, creating structural adversarial relationships with policyholders.

SOLUTION OVERVIEW

ShieldFi: Insurance Rebuilt On-Chain



ShieldFi replaces legacy intermediaries with a fully on-chain, community-governed insurance protocol — transparent by design, automated by default.

Decentralised Underwriting

Liquidity providers collectively back coverage via ERC4626 vaults.

Automated Claims

Chainlink oracles trigger payouts without human gatekeepers.

DAO Governance

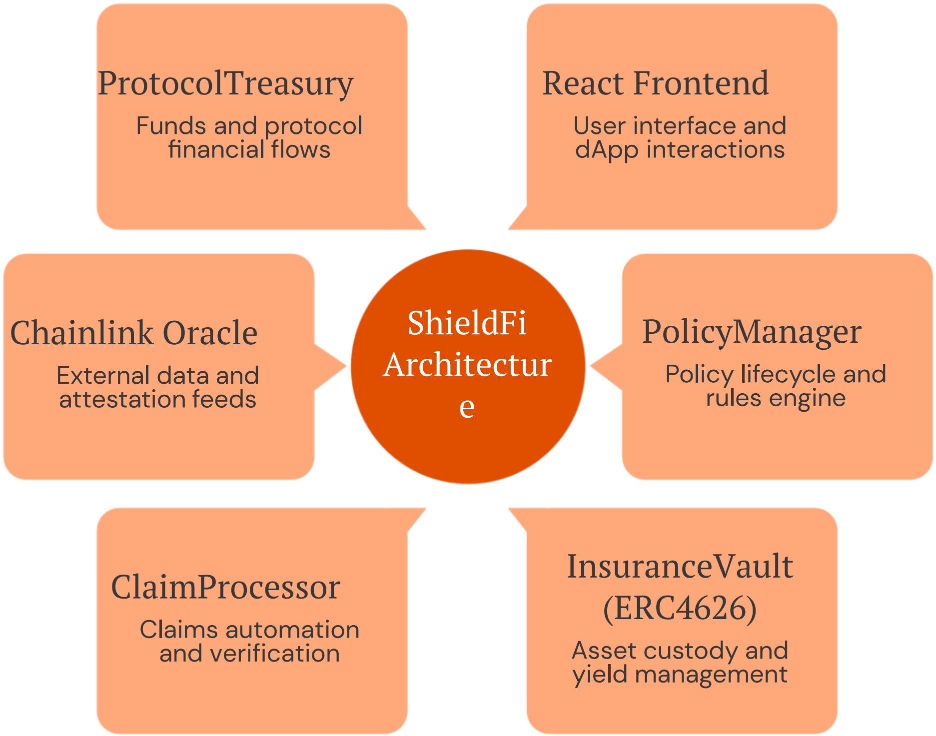
Token-weighted proposals govern protocol parameters on-chain.

Transparent Analytics

The Graph indexes all events for real-time, trustless analytics.

SYSTEM ARCHITECTURE

Protocol Component Map



ShieldFi's architecture separates concerns cleanly across vault management, policy lifecycle, claims automation, DAO governance, and data indexing — each component communicating through well-defined interfaces.

ERC4626 VAULT

InsuranceVault — Tokenised Underwriting

The `InsuranceVault` implements the **ERC4626 tokenised vault standard**, converting liquidity provider deposits into transferable vault shares that represent proportional ownership of the underwriting pool.

01

Deposit

LPs deposit USDC; vault mints proportional `shieldUSDC` shares.

02

Underwrite

Pooled liquidity backs active policies, locking collateral for coverage.

03

Earn Yield

Premium payments accrue to the vault, increasing share value over time.

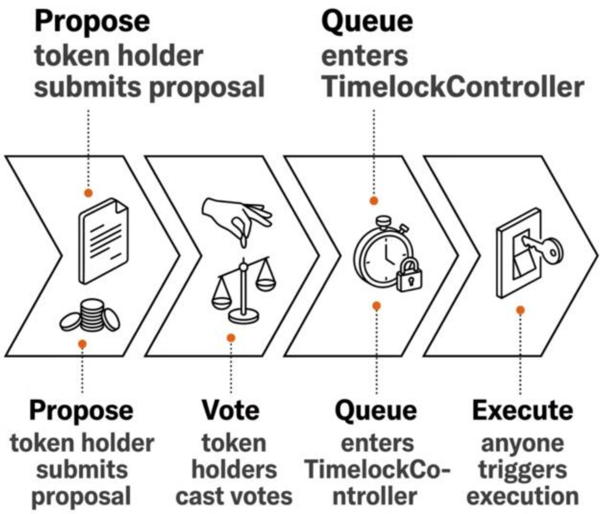
04

Redeem

LPs burn shares to withdraw proportional assets plus accrued yield.

On-Chain Governance Lifecycle

ShieldFi's governance stack implements OpenZeppelin's battle-tested trio: **ERC20Votes** for delegation, **Governor** for proposal logic, and **TimelockController** for execution delay — ensuring no instant unilateral protocol changes.



ERC20Votes

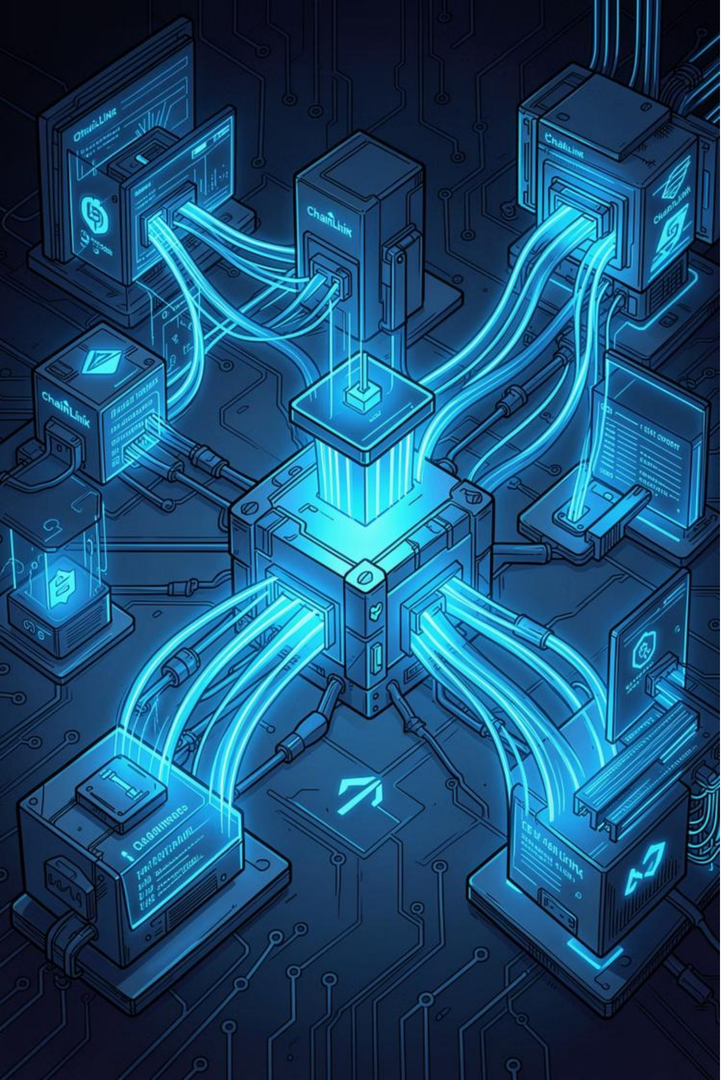
Checkpoint-based voting power with full delegation support.

Governor

Quorum-enforced proposal creation, voting, and queueing logic.

TimelockController

Mandatory execution delay protects against malicious governance attacks.



ORACLE INTEGRATION

Chainlink — Trust-Minimised Data Feeds

Price Feeds

Decentralised aggregator feeds provide tamper-resistant asset prices used to evaluate claim eligibility thresholds.

Stale Price Guards

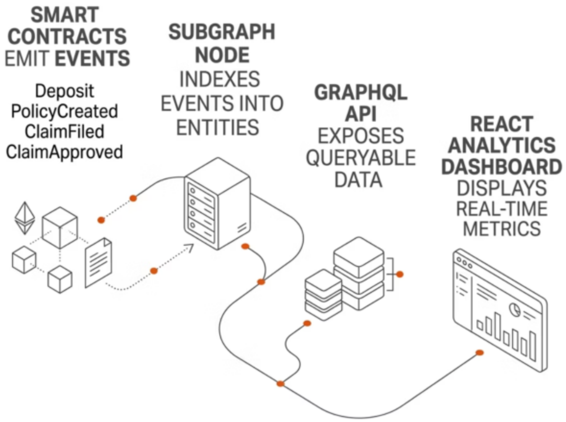
`updatedAt` timestamps are validated on every oracle read — reverting if data exceeds the freshness window.

Automated Triggers

When an oracle-confirmed condition is met, `ClaimProcessor` auto-initiates payout — no human adjudicator required.

THE GRAPH

Subgraph Indexing & Analytics



The ShieldFi subgraph, deployed to **The Graph Studio**, transforms raw Ethereum event logs into structured, queryable entities — enabling trustless, real-time protocol analytics without centralised backends.

Indexed Events

Deposits, withdrawals, policy purchases, and claim lifecycle events.

GraphQL Queries

Typed schema enables efficient filtered queries from the React frontend.

Analytics Dashboard

TVL, active policies, claims ratio, and governance activity — all on-chain verifiable.

TESTING & SECURITY

Hardened by Testing & Static Analysis

241

Tests Passed

Zero failures across all suites.

4

Test Types

Unit, fuzz, invariant, and fork.

100%

Branch Coverage

Critical paths fully covered.

Testing Strategy

- **Unit Tests** — isolated function-level correctness for all contracts.
- **Fuzz Tests** — randomised inputs probing boundary conditions and integer arithmetic.
- **Invariant Tests** — stateful property checks ensuring vault solvency and share consistency.
- **Fork Tests** — mainnet-forked integration tests validating Chainlink and ERC4626 behaviour.

Security Measures

Slither

Static analysis — zero high-severity findings.

ReentrancyGuard

Applied to all state-mutating external calls.

AccessControl

Role-based permissions across all privileged functions.

SafeERC20

Protects against non-compliant token transfer edge cases.

CONCLUSION

ShieldFi — A Production-Grade DeFi Insurance Protocol

ShieldFi demonstrates that decentralised insurance is not only technically viable but architecturally superior — combining ERC4626 vaults, DAO governance, oracle automation, and trustless analytics into a cohesive, auditable system.

🏗️ Architecture

9-component modular protocol deployed on Ethereum Sepolia with verified contracts.

🗳️ Governance

Full DAO lifecycle with ERC20Votes, Governor, and TimelockController.

🧪 Testing

241 passing tests across unit, fuzz, invariant, and fork suites. Zero failures.

🚀 Roadmap

L2 deployment, NFT policy positions, dynamic pricing, and mobile frontend ahead.